Factory Size, Machinery and Output per Worker: Smith, Marx and the Marginalists in Indian Registered Manufacturing, 2021–22 to 2023–24

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Abstract

How do Adam Smith's theory on division of labour, Karl Marx's analysis of machinery and the marginalists' theory on capital productivity explain how large factories produce higher output per worker than small ones? This report uses unit level Annual Survey of Industries (ASI) data over the periods of 2021–22, 2022–23 and 2023–24, 167,886 observations in total. Median gross output per person engaged in large factories (200+ persons) comes out to be around 1.7 times of that in small plants (<50). Capital per worker increases with increase in size, and adding that lowers the estimated size coefficient by almost 30 percent. A 10 percent rise in net fixed capital per worker is related to around 3.7 per cent higher output per worker, within industry and year; the corresponding figure for plant and machinery appears to be about 3.0 per cent. A size effect survives these controls, but it builds up below about 200 persons and is statistically flat above. Large factories pay out a smaller share of value added as emoluments in aggregate (30 per cent against 35–39 per cent), but once industry and year are held fixed the share in large factories is not significantly lower than in small ones (it is 1.7 percentage points higher, not significant at the 5 per cent level), so the data do not support the distributive prediction drawn from Marx. We find no evidence of diminishing returns to capital intensity in this cross-section. All results are associations, not causal effects.

1. Introduction

The Indian registered manufacturing sector appears to be highly unequal in terms of productivity. While a few large plants account for most of the employment and output generated, a long tail of small factories operate with very little capital and low output per head. The Classical thought of political economy offers an explanation for such observations, and so do the marginalists who came after. These two schools of thought disagree on how productivity is defined, its origin, and its implications for workers.

The research question thus arises: *How do Smith's division of labour, Marx's analysis of machinery, and the marginalist theory of capital productivity explain differences in output per worker between large and small factories?* This report has three goals; Firstly, to translate each of these approaches into a testable hypothesis. Secondly, to measure how output per worker varies with factory size and machinery in the dataset. Thirdly, to clarify the parts of each of these approaches that the data can and cannot properly justify.

The empirical approach is based on descriptive statistics and regression. We build a factory level clean dataset from the raw ASI blocks, compare the size classes, and estimate log-lin models of output per worker engaged on factory size and capital intensity, with year and detailed industry fixed effects. It is to be noted that each of the traditions are deliberately treated as competing sources of hypotheses and none is assumed to be correct from the get go.

2. Theoretical propositions

2.1 Smith: division of labour

In *The Wealth of Nations* (1776, Book I, ch. 1–3), Smith attributes the growth of productive power to the division of labour, I.e. by breaking down a production process into several smaller operations, each of them performed by different workers. According to Smith, higher productivity by this process is achieved through three channels: greater dexterity in each of the tasks, reduced time spent in switching between tasks, and the invention of machines that improve labour efficiency. However, he also states that this idea is limited by the extent of the market. A larger factory producing for a larger market can subdivide the workload further and further. This yields the first hypothesis.

H1 (Smith). Output per worker increases as the establishment is scaled, and the effect of this scale remains, even after taking capital per worker and industry into consideration. A positive coefficient on the factory size signifies the claim empirically. Since Smith lists machinery as one of the channels, a scale effect that vanishes once capital is regulated would be consistent with Smith's third channel but would leave dexterity and time saved unmarked.

2.2 Marx: machinery and the labour process

In *Capital*, Volume I (1867, ch. 13–15), Marx defines machinery as a mechanism that makes the labour process more efficient, increasing productive capacity, and placing the pace and organisation of work as being more capital controlled than labour determined. As per Marx, this increase in productivity gain is not a gain shared automatically with the labourers: machinery raises the relative surplus by cheapening the goods workers consume and hence reducing the labour time needed to reproduce wages.

H2a (Marx, productive side). Factories with more machines per worker yield higher output per worker.

H2b (Marx, distributive side). If productivity gains are reflected mainly by capital, the share of value added paid to employees as compensation should be lower in the larger factories with more machinery. Data on Productivity alone cannot prove exploitation or imply changes in welfare; H2b can be analysed using the ASI's cost data, but only to a rough extent.

2.3 Marginalist theory: capital productivity

Marginalist production theory (Clark, Marshall and their successors) explains output as a function of the factors of productions, mainly labour and capital. As per the marginalists, each input is paid according to the value of its marginal product. Technology remaining constant, higher capital per worker increases output per worker, but at a diminishing rate.

H3a. There exists positive association between capital intensity and labour productivity.

H3b. This association is weakened as capital intensity rises due to diminishing marginal returns, that implies a negative coefficient on the squared term of log capital intensity.

It is noteworthy that there exists a caution that is applicable in this case. The Cambridge capital controversy (Robinson 1953–54; Samuelson 1966) questioned if capital can be measured as an aggregate singular value independent of the distribution or prices. Our measure in this report is the book value of fixed assets. Estimating an aggregate production function on it does not assume that the debate is settled, and this report does not claim to interpret the corresponding coefficient as a pure technical elasticity.

3. Empirical data and methods

3.1 Data

For the analysis, unit level ASI data is used over the periods of 2021–22, 2022–23 and 2023–24, obtained as CSV files from MoSPI's microdata site. The ASI is an annual survey of registered factories, organised into blocks, symbolised with an alphabet. Each year consists of ten blocks (A to J). Block A contains identification, status and the sampling multiplier. Block C contains fixed assets, Block D has working capital, Block E records employment, whereas Blocks F and G collects other expenses and other output. Finally, Blocks H and I contain inputs, and Block J, products. All monetary values are in rupees. Block A consists of one row per factory. The other blocks comprise one row per factory per item, and hence they were reduced to the relevant total row before merging.

3.2 Variable construction

Definitions follow the formulas printed in the ASI 2023–24 schedule (scrutiny items 7 to 9).

Variable	Construction
Number of Persons engaged, L	Average number of persons worked, total employees row in Block E
Output, Y	Net sale of products (Block J total, gross sale value minus GST, excise and other distributive expenses, plus subsidies) + change in finished goods (Block D) + other output items (Block G items 1–4, 6–8, 11) + R&D expenses (F7)
Input	Other expenses (Block F items 1–4, 6–8, 11) + total indigenous inputs (Block H, item 23) + total imported inputs (Block I, item 7)
Gross value added, GVA	Output minus Input
Capital, K	Closing net (depreciated) value of total fixed assets, Block C
Machinery, M	Closing net value of plant and machinery, Block C
Emoluments	Wages and salaries of all employees (Block E) + bonus + provident fund contributions + welfare expenses (Block A)
Labour productivity	LP = Y / L
Value added per person	VAP = GVA / L
Capital intensity	CI = K / L
Machinery intensity	MI = M / L

The treatment of R&D expenditure follows the schedule. The 2023–24 scrutiny formulas have included Block F item 7 (R&D) in both total input and total output, hence cancelling it out in gross value added and only raising the level of gross output, by around 0.07 percent of weighted output. Dropping F7 from the output has no effect on the baseline coefficients (0.105 and 0.368). The formulas, despite being taken from the 2023–24 schedule, was applied to all three years. The earlier schedules are in the MoSPI catalogue but have not been compared line by line; the three data files share the same structure of the

block and variables, and to examine the formulas, the published aggregates have been reproduced. This way, weighted gross value added computed comes out to be Rs 20.5 lakh crore in 2021–22 and Rs 21.9 lakh crore in 2022–23, against the published totals of about Rs 20.5 and 22.0 lakh crore. Weighted persons engaged are 17,154,999 and 18,423,375, vis-a-vis published totals of 17,215,350 and 18,494,962 (differences of 0.35 and 0.39 per cent). Small gaps are expected, since the tabulation procedures published differ in detail, as in the 2021–22 data note says that its tabulation also uses data extracted from the previous year's survey.

There are two comparability issues that required care. Firstly, Block E off the 2023-24 data adds a row for transgender workers, so the row numbers of the totals shift by one (as a result total employees is row 9 in the first two years and row numbered 10 in 2023–24). Rows are mapped by year. Secondly, the raw gross-value columns of Block C are nil for about 44 to 45 per cent of open factories reporting a positive net value, and in the cases we inspected the closing gross value was inconsistent with opening gross value plus additions. Net values were internally consistent, hence all capital measures use net values and it was not feasible to run a gross-capital robustness check.

3.3 Merging, sample and identifiers

All blocks were joined one-to-one on the factory serial number (DSL) within each year, and the row count was checked after every join. The three years were then combined, giving us 205,651 factory-year records. Units are retained with status code 1 (open), a positive multiplier, at least one person engaged, positive output and positive net fixed capital.

Step	2021–22	2022–23	2023–24	Total
All Block A records	68,548	68,462	68,641	205,651
Status 1 (open)	55,087	56,230	57,274	168,591
$L \geq 1$	55,086	56,229	57,274	168,589
Output > 0	55,053	56,198	57,231	168,482
Net fixed capital > 0 (final sample)	54,843	55,999	57,044	167,886

Note that the serial number is not a panel identifier. Amongst serial numbers appearing in consecutive years, only 1.4 percent (2021–22 to 2022–23) and 4.2 percent (2022–23 to 2023–24) share both state and five-digit industry. Hence, we treat the data as three repeated cross-section data and do not attempt a panel estimation at the factory level. The five-digit NIC-2008 code denotes the industry and the leading zero is dropped in the raw file; restored by integer division.

3.4 Factory size

Sizes are classified in advance on the basis of persons engaged: small (<50), medium (50–199) and large (200+). This was done before the results were obtained. Small, medium and large factories are 43.3, 29.3 and 27.4 percent of the sample respectively. Weighted by the ASI multiplier, they employ 12.0, 21.7 and 66.3 percent of the workforce respectively.

3.5 Weights, prices and estimation

Block A provides a multiplier. Census scheme units have a multiplier of one while sampled units have larger values. Descriptive aggregates such as shares or per-person totals use the multiplier. Medians are not weighted. The baseline regression is unweighted, and we report a multiplier weighted version to examine the robustness.

The values are nominal and not deflated. In the pooled regressions, year fixed effects absorb the common prices, and the yearly regressions remain unchanged by it. Nominal levels should not be compared across years without further adjustment.

The baseline model is formulated as

$$\ln(LP_i) = \beta 0 + \beta 1 \ln(CI_i) + \beta 2 \ln(L_i) + year_t + industry_j + \varepsilon_i$$

This analyses the association between output per worker, capital intensity and factory size. The regressand as well as the regressors are in terms of natural logarithms, thus allowing the estimated coefficients to be interpreted as approximate elasticities. This formulation includes fixed effects over years as well as industry effects at the three digit NIC level. Standard errors are clustered by four digit NIC (168 clusters). Variants add state effects, replace CI by MI, use ln(VAP) on the positive-GVA subsample, use size-class dummies, add an interaction, a squared term and a kink at 200 persons, and estimate an equivalent specification using logarithms of output, capital and employment for interpretation. A final regression relates the emoluments share of value added to size class, with the same fixed effects. All estimation is by ordinary least squares.

The coefficient on ln L has a specific meaning. If output follows Y = A K^a L^b, then ln(Y/L) = ln A + a ln(K/L) + (a + b − 1) ln L, so the coefficient on ln L equals returns to scale minus one: zero under constant returns, positive under increasing returns. Because L is a denominator in both the dependent variable and capital intensity, measurement error in L would bias the ln L coefficient downward and the capital coefficient upward. The levels equation (ln Y on ln K and ln L) is algebraically the same model, reported for interpretation and not as an independent check. Output is gross of intermediate inputs, so the capital coefficient is not capital's share of value added.

4. Findings

4.1 Descriptive evidence

The following table reports medians by size class and year, in units of ‘Rs lakh per person engaged’ (nominal).

Table 1. Medians by factory size and year (N and median L are counts; the other columns are Rs lakh per person engaged, nominal)

Year	Size	N	Median L	Output	GVA	Capital	Machinery	Emoluments
2021–22	Small	24,477	15	19.76	3.44	4.19	1.24	1.63
2021–22	Medium	15,988	109	25.32	4.45	4.58	1.61	2.39
2021–22	Large	14,378	403	33.94	6.26	7.06	2.97	3.01

Year	Size	N	Median L	Output	GVA	Capital	Machinery	Emoluments
2022–23	Small	24,031	15	22.09	3.71	4.54	1.33	1.74
2022–23	Medium	16,612	109	27.62	4.74	4.93	1.69	2.55
2022–23	Large	15,356	406	37.18	6.48	7.21	3.03	3.20
2023–24	Small	24,198	15	22.36	3.90	4.95	1.53	1.81
2023–24	Medium	16,662	109	28.14	5.06	5.24	1.82	2.73
2023–24	Large	16,184	412	37.44	6.95	7.62	3.23	3.40

Each measure is monotonic increasing with respect to size in each time period. Pooled over years, median output per person comes out to be Rs 21.4 lakh in small, 27.0 lakh in medium and 36.2 lakh in large factories, making the latter about 1.7 times as productive at the median. Median capital per person engaged comes out to be around 1.6 times higher and median machinery per person about 2.3 times higher in large than in small factories. The median factory's nominal output per person increased from Rs 25.0 lakh in 2021–22 to 27.8 lakh in 2022–23 and finally to 28.4 lakh in 2023–24; in the weighted aggregate it rose from 69.6 to 78.6 and then was flat at 78.5 through these 3 time periods. These are the nominal changes and include price inflation.

This next table uses the multiplier to describe the factory population. Large factories account for two-thirds of employment generated and almost three fourth of output produced.

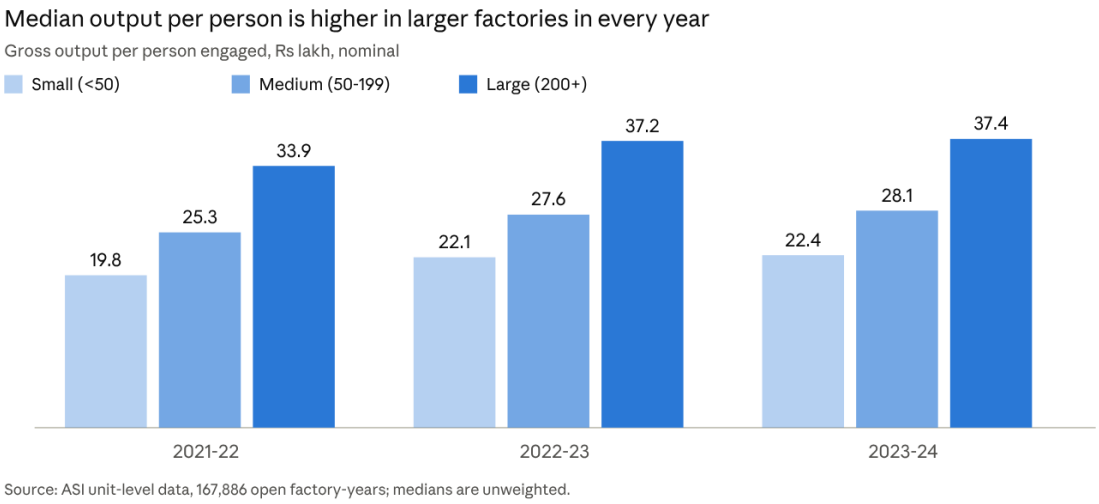
Table 2. Pooled weighted aggregates by size (Rs lakh per person engaged, nominal)

Size	Share of workforce	Share of output	GVA / person	Capital / person	Emoluments / person	Emoluments as share of GVA
Small (<50)	12.0%	9.8%	6.91	14.42	2.45	35.4%
Medium (50–199)	21.7%	16.3%	7.82	12.25	3.06	39.2%
Large (200+)	66.3%	73.8%	14.56	27.13	4.42	30.4%

Aggregate means are pulled up by a few very capital-intensive plants, which is why medium factories show a lower average capital per person than small plants even though their median is higher. The distributions are strongly skewed to the right, so medians are the safer measure of central tendency.

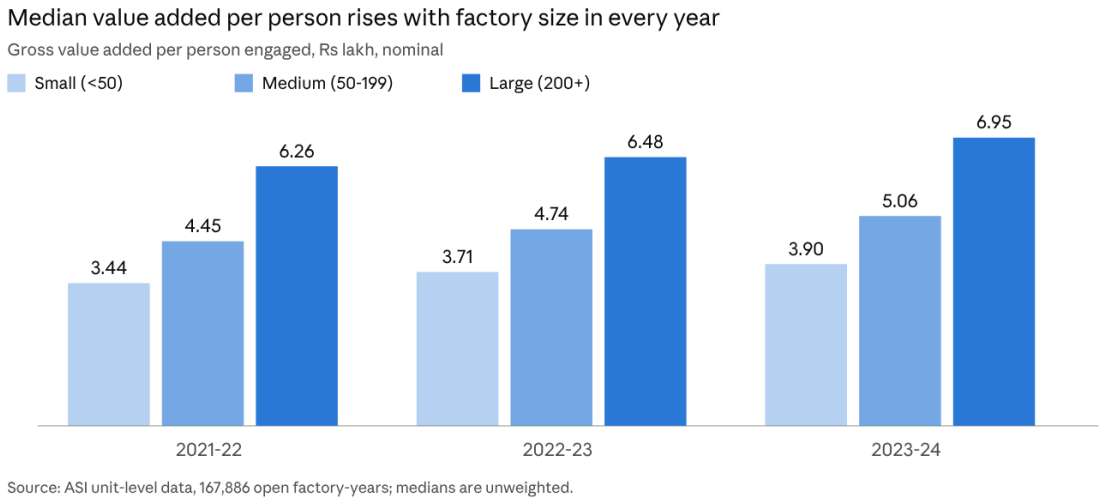
As given below, Figures 1 to 3 plot the medians of Table 1: the size ordering is the same for every measure in every year. Figure 4 shows the labour productivity given by output per person by year, for the median factory and for the weighted aggregate, both being nominal.

Figure 1. Median output per person, by factory size and year



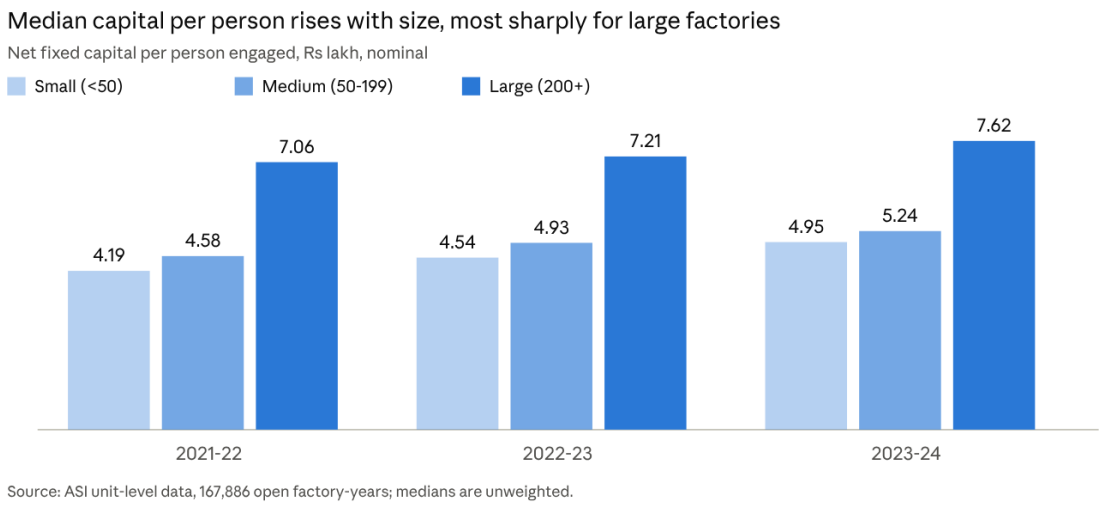
ASI unit-level data · medians, nominal

Figure 2. Median value added per person, by factory size and year



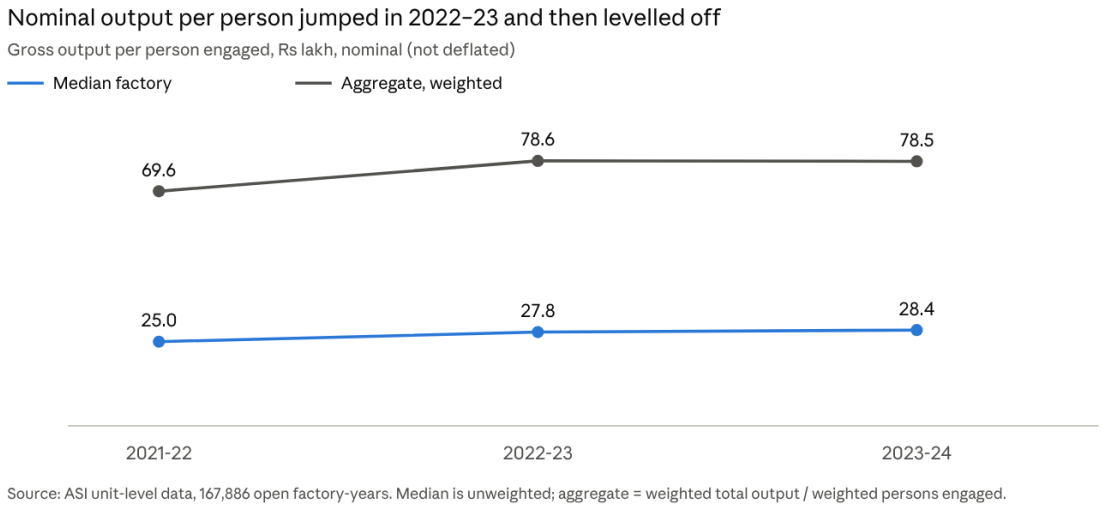
ASI unit-level data · medians, nominal

Figure 3. Median net fixed capital per person, by factory size and year



ASI unit-level data · medians, nominal

Figure 4. Output per person by year



ASI unit-level data · nominal, not deflated

4.2 Regression evidence

The third table on this report explains the main models. Standard errors, clustered by four-digit NIC, are in parentheses. The dependent variable is ln(output per person engaged) unless noted.

Table 3. Regression results

Model	ln L	ln CI	ln MI	R^2	N
(1) Size only, year FE	0.148 (0.022)			0.034	167,886
(2) + capital intensity, year FE	0.104 (0.017)	0.415 (0.033)		0.327	167,886
(3) Baseline: + NIC-3 FE	0.105 (0.012)	0.368 (0.021)		0.439	167,886
(4) + state FE	0.095 (0.010)	0.347 (0.016)		0.463	167,886
(5) Baseline, multiplier-weighted	0.101 (0.016)	0.364 (0.020)		0.395	167,886
(6) Machinery replaces capital	0.082 (0.014)		0.298 (0.019)	0.399	164,435
(6b) Capital and machinery together	0.090 (0.012)	0.311 (0.017)	0.092 (0.009)	0.441	164,435
(7) ln(GVA per person), GVA > 0	0.134 (0.010)	0.299 (0.016)		0.419	162,322

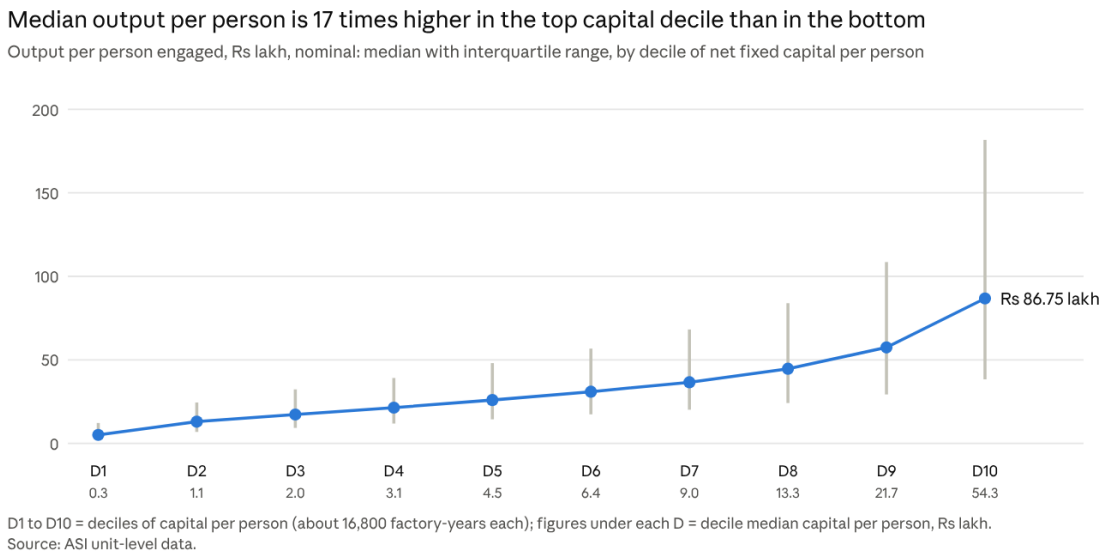
Fixed effects: year in all models; three-digit NIC in models 3 to 7; state added in model 4. Model 6 uses factories with positive machinery value.

Size (H1). Factory Size alone explains very negligible amount, with an R^2 of 0.03 and an elasticity of 0.15. Adding capital intensity further cuts the size coefficient from 0.148 to 0.104. Hence adding capital per worker lowers the estimated size coefficient by almost 30 percent. Capital intensity statistically accounts for a portion of the association between factory size and productivity; this comparison however does not recognise any causal mechanism. All else remains unchanged. At the baseline, doubling the number of persons engaged is associated with about 7.5 percent increased output per worker (2^0.105 − 1), within industry and year. Using size classes instead as given in Table 5, medium factories have around 19 percent higher output per worker than small factories having the same capital intensity and industry, and large factories around 36 percent more.

Capital and machinery (H2a, H3a). Capital intensity has a strong and precise relation with productivity: a coefficient of 0.37 in the baseline regression. Machinery intensity has a coefficient of 0.30 by itself, and when both are included capital intensity dominates (0.31 vis-a-vis 0.09). Capital intensity and machinery intensity come out to be highly correlated (0.83 in logs), and hence the data cannot clearly separate machinery from rest of the fixed assets like buildings and land. Replacing gross output by value added per person makes some difference, as the size coefficient increases to 0.13 and that of capital's decreases to 0.30.

Figure 5 shows this same relation without any model. Output per person steadily increases across deciles of capital per person, but the spread between each decile is wide enough, so capital intensity fails to explain most of the variation in productivity: R^2 is 0.03 with size only and 0.33 once capital intensity is included.

Figure 5. Output per person by capital-intensity decile



ASI unit-level data · 167,886 factory-years · Rs lakh per person engaged, nominal

Robustness. The baseline stays stable across the three years (size 0.105, 0.102, 0.106; capital 0.366, 0.370, 0.368 for 2021–22, 2022–23 and 2023–24 respectively). Cutting down the top and bottom one percent of output per person yields 0.105 and 0.347; restricting to census-scheme factories yields 0.116 and 0.371; and measuring labour by workers instead of persons engaged gives a smaller size coefficient (0.046) and a bigger capital coefficient (0.411). Conditional on size and industry, nominal output per

person is 7.3 log points higher in 2022–23 and 5.7 log points higher in 2023–24 than in 2021–22. Since we do not deflate, this does not prove to be a real productivity change.

4.3 Where the size premium sits, and diminishing returns

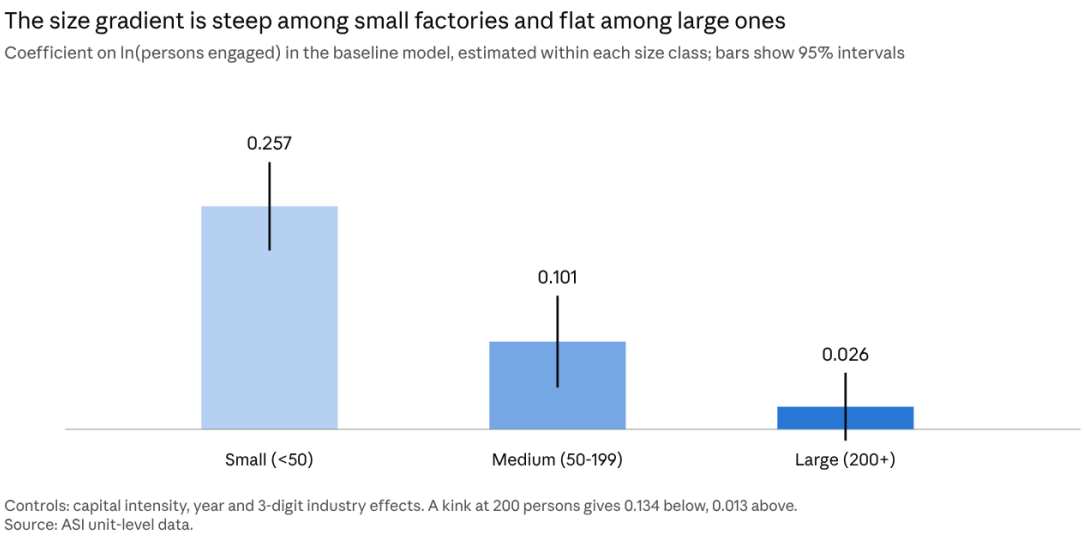
This next table splits the sample by size class and fits the baseline within each classification.

Table 4. Baseline model within size classes

Sample	ln L	ln CI	N
Small (<50)	0.257 (0.026)	0.326 (0.014)	72,706
Medium (50–199)	0.101 (0.027)	0.395 (0.023)	49,262
Large (200+)	0.026 (0.020)	0.362 (0.038)	45,918

The conditional elasticity of factory size decreases from 0.26 amongst small factories to 0.10 amongst medium ones and 0.03, statistically indistinguishable from zero, among the large ones, while the capital coefficient stays in the middle of 0.33 and 0.40 as shown in Figure 6. A single model with a kink at 200 persons agrees: the size slope is 0.134 (0.017) below 200 persons and 0.013 (0.028) above. The size class dummy variables in Table 5 show that large factories still lie above medium plants on an average (36 percent vis-a-vis 19 per cent above small factories at the same capital intensity), so the premium is a step up through the small-to-medium range that stops accumulating above about 200 persons, not an absence of any large-factory advantage. The pooled interaction of ln CI with ln L is positive but small (0.017, standard error 0.008).

Figure 6. Size elasticity of productivity within each size class



ASI unit-level data · baseline model estimated separately by size class

For H3b hypothesis we add a squared term in log capital intensity, centred at its average. Its coefficient is positive (0.015 with standard error 0.002). The elasticity of output per worker with respect to capital

intensity thus implied increases from 0.35 at the 10th percentile of capital intensity to 0.42 at the median and 0.47 at the 90th percentile, which is the opposite of diminishing returns. This should not be interpreted as technical increasing returns, factories at different points of the distribution have varying technique, product mix and capital age, and book values do not consider these constant. The test shows that it is not possible for a positive capital coefficient alone to establish diminishing returns, and that the extra test does not support that here.

Table 5. Additional specifications (year and three-digit NIC effects throughout; standard errors in parentheses, clustered by four-digit NIC)

Specification	Key estimates	N
Size classes (reference: small)	medium 0.172 (0.051); large 0.309 (0.037); ln CI 0.371 (0.019)	167,886
Interaction of ln CI and ln L	ln L −0.116 (0.101); ln CI 0.300 (0.025); interaction 0.017 (0.008)	167,886
Kink at 200 persons	ln L below 0.134 (0.017); above 0.013 (0.028); ln CI 0.370 (0.021)	167,886
Levels: ln output on ln K and ln L	ln K 0.368 (0.021); ln L 0.737 (0.027); sum 1.105	167,886
Squared ln CI (centred)	ln L 0.100 (0.011); ln CI 0.411 (0.023); square 0.015 (0.002)	167,886
Emoluments / value added, size classes	medium +0.054 (0.009); large +0.017 (0.011)	160,698
Emoluments / output, size classes	medium +0.017 (0.008); large +0.003 (0.005)	166,207

The two share regressions use the size-class dummies relative to small factories; the first keeps factories with positive value added, and both drop the top 1 per cent of the ratio. In levels, output rises 0.37 per unit of ln capital and 0.74 per unit of ln persons engaged, a sum of 1.105: mild increasing returns to scale on gross output. This restates the 0.105 size coefficient rather than testing it independently.

4.4 Relating the findings to the three traditions

Smith. The findings of the analysis fits Smith's ideas in a qualified manner. Larger factories have higher productivity, and a size premium of around 7 to 8 percent per doubling of employment, I.e. persons engaged remains even after accounting for controls for capital and industry. This is consistent with the idea of division of labour beyond mechanisation. However, this premium is concentrated amongst plants with below about 200 persons and the relationship between size and productivity weakens above that level. This points to Smith's idea of limit of the extent of the market; On the other hand it is that factories of that scale may have already run out of the organisational gains available within their industry, or that the size measure is incapable to capture other differences. Since specialisation, dexterity or time saving is not directly measured in the data analysed, these effects cannot be differentiated.

Marx. Marx's idea about the productive role of machinery is supported by the findings: factories with more capital and more machinery per worker tend to have higher output per person. Adding capital per worker also lowers the estimated size coefficient by almost 30 percent. On the other hand, the

distributive side is not supported by the evidence. At an aggregate level, large factories tend to produce 2.1 times as much value added per person as compared to small plants (14.56 vis-a-vis 6.91 lakh) albeit paying only 1.8 times as much in employee emoluments per worker (4.42 vis-a-vis 2.45 lakh), hence emoluments account for 30.4 percent of value added in larger factories vis-a-vis 35.4 per cent in small and 39.2 percent in medium plants. To be noted that this difference does not show up within industries. Within the same three-digit industry and in a given year, the emoluments share of value added is 5.4 percentage points more in medium plants and 1.7 points higher in large plants than in small factories, and the difference for large plants is not significant at 5 percent level, and the emolument-output share moves by less than 2 percentage points. Hence, the aggregate difference reflects industry mix and the weight of a few highly capital-intensive plants, instead of a within industry pattern for larger, more mechanised factories to pay a smaller share in employee emoluments. Hence, there is no empirical support for H2b. This test is only a rough ideation of Marx's argument on distribution. Value added is measured as gross of depreciation, including interest, rent, profit and managerial pay, while the wage measure covers all employees, even supervisors. However, in this study surplus value or rate of exploitation could not be analysed since we do not observe working hours, working intensity or working conditions, and hence it cannot be said whether workers in large factories are better off or worse off.

Marginalists. The positive and relatively steady relationship between capital per worker and output per worker as estimated to be about 0.3 to 0.4 matches the hypothesis H3a and is the clearest finding of this report. Although interpreting the coefficient as a marginal product would imply competitive factor market assumptions which is not examined here. The increasing instead of falling marginal elasticity also does not support the diminishing returns assumption of H3b. Along with the issues in measuring capital as said in section 2.3, this points towards treating the capital coefficient as a robust empirical relationship rather than the marginal product of capital.

4.5 Limitations and alternative explanations

The findings point out relationships from three repeated cross section data and hence does not provide any causal interpretation. Reverse causation can come into the picture since productive factories are able to spend in more capital and hence grow bigger. Some other factors, such as worker skill, management quality, product mix, ownership of firm, location and utilisation of capacity, are not observed yet would be correlated with both size and capital.

On the other hand, capital is measured as book value minus depreciation, hence incorporating both effects of both age and price of assets; the raw gross values are not usable. Output is measured as gross of intermediate inputs, and hence the Labour productivity measured by output per person partially reflect gaps in the input intensity. That is why value added is examined as a check of robustness. Between 3.0 and 3.6 percent of open factories are found to show negative value added, and hence are excluded from the value added models. The unit of this analysis is the factory rather than the firm, so multi plant setups are treated as separate observations. Employment is measured as head count and does not consider work hours or status of the contract. Since employment is also the denominator of the regressand as well as capital intensity, measurement error in the same would bias the size coefficient downward and the capital coefficient upward. All monetary values are nominal and have not been adjusted for inflation, limiting comparisons of levels across years. This report uses ASI's own status code and multiplier but does not verify the survey sampling design beyond what is stated in the documentation.

5. Conclusion

Large Indian factories generate significantly more output per worker than small plants. Adding net fixed capital per worker lowers the estimated size coefficient by about 30 percent, from 0.148 to 0.104. This suggests that capital intensity statistically accounts for a portion of the association between size and productivity, although not establishing any causal relationship. The rest persists even after conditioning on industry. There exists a robust positive relationship between Capital intensity and productivity in every year and under every possible specification tried. The size premium itself is concentrated amongst small and medium factories and weakens above the 200 person mark. Larger factories tend to distribute a smaller share of value added as employee emoluments in aggregate, but not at a statistically significantly level within industries.

For each of the three traditions, the data support Smith's claim to some bounded extent, it agrees with Marxist ideas on the productive side while giving no substantial backing to the distributive side, and it supports the marginalists' association of capital with output without backing up diminishing returns. The analysis cannot establish any causation, and neither can it distinguish the effects of dexterity, time saved and mechanisation, or imply anything about workers' welfare or rate of exploitation.

Future research on the same could use a factory level panel data source, so as to be able to distinguish changes within factories from fixed differences between factories. Deflating values using industry specific price index can help in studying changes over time. Data on working hours and differences in wages on the basis of skill level could help in understanding labour conditions. Also, future studies can analyse whether the weakening of the size to productivity association above the 200 person mark reflects any technological barriers or constraints.

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